Closing the Rank-Four Gap for f-Factors in Balanced Hypergraphs

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Abstract

We study exact incidence-degree problems in balanced hypergraphs. Previous work established polynomial solvability for rank at most three and NP-completeness for non-uniform balanced hypergraphs using a rank-five construction, leaving rank four open. We close this gap. We give a simple rank-four selector gadget and prove that it is balanced. A reduction from bounded-occurrence three-dimensional matching shows that both the f-factor problem and the perfect f-matching problem are NP-complete even when the maximum degree is at most ten and $\max_v f(v) \leq 5$. Applying the standard vertex-expansion operation yields NP-completeness of ordinary Perfect Matching for non-uniform Mengerian hypergraphs of rank at most four and maximum degree at most fifteen. Since Mengerian hypergraphs are ideal, the same instances give an ideal-hypergraph corollary. Thus the complexity dichotomy by rank bound for non-uniform balanced f-factor and perfect f-matching problems is polynomial for bounds at most three and NP-complete already for bound four.

1 Introduction

Let $H = (V, E)$ be a finite hypergraph. For a function $f : V \to \mathbb{Z}_{\geq 0}$, an f-factor is a subset of edges in which vertex v has incidence degree exactly $f(v)$. The related perfect f-matching problem permits nonnegative integer multiplicities on edges. These are natural exact-covering problems, and their complexity depends strongly on the incidence structure of H.

Balanced hypergraphs are defined by the absence of strong odd cycles. Their incidence matrices have strong integrality properties; in particular, balanced hypergraphs of rank at most three have totally unimodular incidence matrices. Beckenbach and Scheidweiler established polynomial solvability in that range and proved NP-completeness for non-uniform balanced hypergraphs with a rank-five reduction [1]. Their paper explicitly identified rank four as the remaining case; the same boundary is recorded in the more detailed dissertation account [2].

We prove that rank four is already hard. The central gadget replaces the rank-five selector edge used previously. For every source triple, it has a center c, helpers $h_1, \ldots, h_4$, auxiliary vertices r_1, r_2, four fallback edges, two rank-four selector edges, one rank-three synchronizing edge, and three source ports. Two equations at r_1, r_2 synchronize the selector edges; the center equation then forces either zero or all three ports. The only delicate point is global balancedness, which follows from a case analysis of strong cycles.

Our results are as follows.

Theorem 1.1 (Balanced rank-four factors)**.** *The f-factor problem and the perfect f-matching problem are NP-complete for non-uniform balanced hypergraphs of rank at most four. The hardness holds for instances satisfying*

$$\Delta(H) \leq 10, \qquad \max_{v \in V} f(v) \leq 5.$$

Theorem 1.2 (Mengerian rank-four perfect matching)**.** *Perfect Matching is NP-complete for non-uniform Mengerian hypergraphs of rank at most four and maximum degree at most fifteen.*

The construction is non-uniform. We make no claim about uniform rank-four hypergraphs, nor about counting versions of the problems.

2 Preliminaries

We use simple hypergraphs: distinct edges are distinct subsets of V, and no repeated edges occur. The degree of v is the number of edges containing v, and $\Delta(H)$ is the maximum degree. The rank is $\max_{e\in E}|e|$.

Definition 2.1 (Strong cycle and balancedness)**.** A strong cycle of length $k \geq 2$ is a cyclic sequence of distinct vertices $v_1, \ldots, v_k$ and distinct edges $e_1, \ldots, e_k$ such that

$$e_i \cap \{v_1, \ldots, v_k\} = \{v_i, v_{i+1}\}$$

for every i, with indices modulo k. A hypergraph is balanced if it has no strong cycle of odd length.

Definition 2.2 (f-factors and perfect f-matchings)**.** An f-factor is a subset $F \subseteq E$ satisfying

$$|\{e \in F : v \in e\}| = f(v) \quad (v \in V).$$

A perfect f-matching is a vector $x \in \mathbb{Z}^E_{\geq 0}$ satisfying

$$\sum_{e \ni v} x(e) = f(v) \quad (v \in V).$$

When $f \equiv 1$, the latter is an ordinary perfect matching.

Definition 2.3 (Mengerian and ideal hypergraphs)**.** For a hypergraph K, let $\nu(K)$ be its maximum matching size and $\tau(K)$ its minimum vertex-cover size. For a nonnegative integer vector a on $V(K)$, let K^a denote the vertex expansion obtained by replacing v with $a(v)$ clones and replacing every incident edge by all choices of one clone. A hypergraph is Mengerian if

$$\nu(K^a) = \tau(K^a)$$

for every a. It is ideal if its vertex-cover polyhedron

$$\{y \in \mathbb{R}^V_{\geq 0} : \sum_{v\in e} y_v \geq 1 \ (e \in E)\}$$

has only integral vertices.

Balanced hypergraphs are Mengerian, and every Mengerian hypergraph is ideal; these are standard min–max consequences for balanced incidence matrices and their expansions [1, 2].

3 The rank-four construction

We reduce from three-dimensional matching. The source consists of disjoint sets X, Y, Z of equal size and a set $M \subseteq X \times Y \times Z$. A perfect 3-dimensional matching is a set of triples covering every source vertex exactly once. We use the standard bounded-occurrence version in which every source vertex occurs in at most three triples; this version is NP-complete [3].

For each $e = (v_1, v_2, v_3) \in M$, introduce private vertices

$$c_e, h_{e,1}, h_{e,2}, h_{e,3}, h_{e,4}, r_{e,1}, r_{e,2}.$$

The edges of the gadget are

$$\begin{aligned}
U_{e,i} &= \{c_e, h_{e,i}\} && (1 \leq i \leq 4),\\
B_{e,1} &= \{c_e, h_{e,1}, h_{e,2}, r_{e,1}\},\\
B_{e,2} &= \{c_e, h_{e,3}, h_{e,4}, r_{e,2}\},\\
A_e &= \{c_e, r_{e,1}, r_{e,2}\},\\
P_{e,j} &= \{c_e, v_j\} && (1 \leq j \leq 3).
\end{aligned}$$

Set $f(c_e) = 5$ and $f(w) = 1$ for every other vertex. The only shared vertices are the source vertices; all gadget vertices are private.

Lemma 3.1 (Two local modes). *In every f-factor, each gadget is in exactly one of the following modes.*

(i) OFF: A_e and all four $U_{e,i}$ are selected, while no B-edge or source port is selected.

(ii) ON: both B-edges and all three source ports are selected, while A_e and all four U-edges are unselected.

Both modes satisfy all local equations.

Proof. Write $u_i, b_1, b_2, a, p_j \in \{0, 1\}$ for the edge indicators. The r-vertices give

$$b_1 + a = 1, \qquad b_2 + a = 1.$$

Thus $b_1 = b_2 = b$ and $a = 1 - b$. Each helper gives $u_i + b = 1$, hence $u_i = 1 - b$. At the center,

$$4(1 - b) + 2b + (1 - b) + \sum_{j=1}^{3} p_j = 5,$$

so $\sum_j p_j = 3b$. If $b = 0$, all ports are absent; if $b = 1$, all three ports are present. Substitution verifies both modes. □

4 Balancedness

The following lemma is the structural core of the reduction.

Lemma 4.1. *Every strong cycle of length greater than two in the constructed hypergraph uses only size-two edges.*

Proof. The only edges of size greater than two are $B_{e,1}, B_{e,2}, A_e$. Fix a gadget e and suppose first that c_e is a cycle vertex. A non-size-two edge containing c_e can contain at most one other cycle vertex. If that vertex is an $h_{e,i}$, the only other edge incident with it is $U_{e,i}$; the two cycle edges at it therefore return immediately to c_e, producing only a 2-cycle. If the other vertex is $r_{e,1}$ (the other case is symmetric), continuation through A_e either uses $r_{e,2}$ as well, in which case A_e contains the three cycle vertices $c_e, r_{e,1}, r_{e,2}$, contrary to strength, or uses c_e and closes a 2-cycle. The same argument with $B_{e,1}$ applies when the current edge is A_e.

Now suppose c_e is not a cycle vertex. No size-two edge incident with c_e can be a cycle edge, because it contains only one cycle vertex. Thus no U- or P-edge occurs. Among the remaining edges, the private supports intersect only as

$$B_{e,1} \cap A_e = \{r_{e,1}\}, \quad A_e \cap B_{e,2} = \{r_{e,2}\}, \quad B_{e,1} \cap B_{e,2} = \varnothing.$$

They form a path, not a cycle. A two-edge strong cycle is impossible because each pair shares at most one vertex, while a 2-cycle requires two common cycle vertices. Different gadgets have disjoint private vertices. Hence no strong cycle avoiding c_e can use a non-size-two edge.

It follows that every strong cycle of length greater than two consists only of U- and P-edges. These form a bipartite graph with gadget centers on one side and source/helper vertices on the other, so every such cycle is even. Therefore the hypergraph is balanced. □

5 Hardness for balanced hypergraphs

Theorem 5.1. *The construction has an f-factor if and only if the source 3DM instance has a perfect matching.*

Proof. Given a perfect source matching, put its gadgets in ON mode and all others in OFF mode. The local lemma gives the required degree at every private vertex and every center. Each source vertex is in exactly one selected triple, and hence is incident with exactly one selected port.

Conversely, let F be an f-factor. The local lemma assigns each gadget a unique mode. Let M' consist of the source triples whose gadgets are ON. An original source vertex is incident only with port edges and has required degree one. Since an ON gadget selects all three of its ports and an OFF gadget selects none, exactly one triple of M' contains each source vertex. Thus M' is a perfect 3-dimensional matching. □

The construction has seven private vertices and ten edges per source triple, so it is polynomial. Membership in NP follows by checking a proposed edge subset and all incidence degrees. Together with balancedness, this proves NP-hardness and NP-completeness for rank at most four.

Corollary 5.2 (Perfect f-matching). *The same instances give NP-completeness for perfect f-matching.*

Proof. Every target edge contains a vertex with f-value one: a helper on each U-edge, a helper or r-vertex on each B-edge, an r-vertex on A_e, and a source vertex on each port. Consequently, in any perfect f-matching x, every edge multiplicity satisfies $x(e) \leq 1$. Thus x is a $0/1$ vector and is exactly an f-factor. □

For the restricted problem, membership in NP is explicit: a certificate is the $0/1$ incidence vector of the selected edges, and all vertex degrees can be checked in polynomial time. For unrestricted binary-encoded inputs, an integer multiplicity vector is likewise polynomially encoded by its bounded coordinates.

6 Bounded degree and the rank dichotomy

For the bounded-occurrence source, the target degrees are

$$\deg(c_e) = 4 + 2 + 1 + 3 = 10, \quad \deg(h_{e,i}) = 2, \quad \deg(r_{e,i}) = 2,$$

and every source vertex has degree at most three. Therefore $\Delta(H) \leq 10$ and $\max_v f(v) = 5$. This proves the strengthened form of Theorem 1.1.

Combining the present theorem with the prior rank-at-most-three algorithm and rank-five hardness gives the precise rank-bound classification:

rank bound	f-factor and perfect f-matching	status
$r \leq 3$	on balanced hypergraphs of rank at most r	polynomial
$r \geq 4$	already on balanced hypergraphs of rank at most r	NP-complete

The second line means that the class with rank bound r contains a hard rank-four subclass; it does not assert that every input has rank at least four. The construction is non-uniform.

7 Mengerian perfect matching

We use the vertex expansion operation from the prior literature, not as a new notion. The next proposition records the required details.

Proposition 7.1 (Expansion equivalence). *For any hypergraph H and $f : V(H) \to \mathbb{Z}_{\geq 0}$, H has a perfect f-matching if and only if H^f has an ordinary perfect matching. Moreover,* $\operatorname{rank}(H^f) \leq \operatorname{rank}(H)$.

Proof. If x is a perfect f-matching, then for every v the incident occurrences of original edges number exactly $f(v)$. Assign the $f(v)$ clones of v bijectively to these occurrences. Choosing the assigned clone at every endpoint of each occurrence produces expanded edges that are pairwise disjoint and cover all clones. Conversely, project a perfect matching of H^f to original edges and let $x(e)$ be the number of matching edges projecting to e. Counting the clones of each v gives $\sum_{e\ni v} x(e) = f(v)$. Expansion replaces, rather than adds, one vertex at each endpoint, so rank cannot increase. ☐

Proposition 7.2. *If H is balanced, then H^f is Mengerian.*

Proof. Balanced hypergraphs are Mengerian. Let g be any expansion vector on the clones of H^f. If the clones of v in H^f are $v_1, \dots, v_{f(v)}$, define

$$h(v) = \sum_{i=1}^{f(v)} g(v_i).$$

Expanding in two stages gives a natural isomorphism $(H^f)^g \cong H^h$: the final clone of v is identified by the pair (i, j) consisting of an intermediate clone and a clone inside it. Hence matching and cover numbers agree, and

$$\nu((H^f)^g) = \nu(H^h) = \tau(H^h) = \tau((H^f)^g).$$

Thus H^f is Mengerian. ☐

For the bounded-occurrence construction, every edge contains exactly one center, and every center has f-value five. Hence

$$|E(H^f)| = 5|E(H)| = 50|M|, \qquad |V(H^f)| = 3n + 11|M|.$$

The expanded center clones have degree ten. Each helper and each r-vertex has degree $5\cdot 2 = 10$, while each source vertex has degree at most $5 \cdot 3 = 15$. Consequently $\Delta(H^f) \leq 15$, and rank remains at most four.

Proof of Theorem 1.2. Reduce from bounded-occurrence 3DM-3, construct H as above, and output H^f. The preceding propositions show that the output is Mengerian, has rank at most four and maximum degree at most fifteen, and has a perfect matching exactly when the source instance is positive. Perfect Matching is in NP because a selected edge subset can be checked for pairwise disjointness and full vertex coverage in polynomial time. ☐

Corollary 7.3 (Ideal hypergraphs)**.** *Perfect Matching is NP-complete for ideal hypergraphs of rank at most four, even with maximum degree at most fifteen.*

Proof. Every Mengerian hypergraph is ideal, so this is an immediate consequence of Theorem 1.2. ☐

8 A protected selector interpretation

The gadget can be viewed as a rank-reduction replacement for the selector in the earlier rank-five construction. Under the protected attachment pattern used here, the old selector has two local behaviors: four fallbacks with no external incidence, or one large selector with three external incidences. The subsystem B_1, B_2, A has the same external behavior. The equations $b_1 + a = b_2 + a = 1$ synchronize the two rank-four selector edges, and the center requirement five forces the port sum to be either zero or three. The balancedness proof is specific to the stated private incidence pattern; no universal rank-reduction theorem is claimed.

9 Verification note

The proof is independent of computation. Reproducibility details and exact small-instance checks are supplied separately in the repository supplement; they are falsification evidence, not a proof of any theorem.

10 Conclusion

The rank-four gap for non-uniform balanced hypergraph f-factors and perfect f-matchings is closed by a rank-four balanced selector. The same construction, followed by rank-preserving vertex expansion, gives bounded rank hardness for Mengerian and ideal hypergraphs. We deliberately leave uniform rank-four and counting variants outside the scope of this paper.

References

[1] Isabel Beckenbach and Robert Scheidweiler. Perfect f-matchings and f-factors in hypergraphs—a combinatorial approach. *Discrete Mathematics*, 340(10):2499–2506, 2017.

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[3] Michael R. Garey and David S. Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness.* W. H. Freeman, San Francisco, 1979.